Table 1: All measures of the rigidity comparison (test set, 28 patients; network trained with the PET losses, IO with the PET terms and one rigidity penalty at the weight chosen on the validation set). Rigidity of bones: residual of a rigid fit to each bone (minimised by the rigid-fit penalty), Staring energy (minimised by the Staring penalty), and distortion of short (5–15 mm) and long (≥15 mm) within-bone distances (minimised by neither). HD95 over bones and soft-tissue organs; near-bone soft tissue: soft-tissue organs within three voxels of a bone; Staring energy: sum of its properness, orthonormality, and affinity terms on the whole bone; NDV in bone band: in a three-voxel band around the bones. Mean±SD over patients; $^{\dagger}/^{\ddagger}$: significantly better/worse than without rigidity (two-sided Wilcoxon signed-rank test on patient values, Holm-corrected over the three penalties per row, $p < 0.05$).

Measure	No rigidity	Staring (×3)	Staring + erosion (×100)	Per-bone rigid fit (×100)
Alignment				
CT Dice (%)	77.7±4.8	75.7±4.3‡	77.6±4.6	76.1±4.6‡
CT HD95 (mm)	6.96±2.35	7.03±2.35‡	6.90±2.34	7.03±2.35
Bone Dice (%)	78.4±6.4	74.6±5.7‡	77.5±6.2‡	74.5±6.2‡
Bone HD95 (mm)	7.53±3.38	7.68±3.40‡	7.54±3.36	7.79±3.41‡
Soft-tissue Dice (%)	79.9±3.9	79.8±3.9	80.7±3.9†	80.8±4.0†
Soft-tissue HD95 (mm)	6.31±2.94	6.29±2.95	6.18±2.98†	6.18±3.01†
Near-bone soft-tissue Dice (%)	74.1±4.3	73.7±4.2‡	74.9±4.2†	74.8±4.2
Rigidity of bones				
Rigid-fit residual (mm)	2.67±0.52	2.00±0.48†	2.31±0.47†	1.39±0.40†
Staring energy	3.517±2.110	0.272±0.090†	1.803±0.650†	1.419±0.473†
Staring properness	0.182±0.100	0.022±0.009†	0.097±0.033†	0.093±0.027†
Staring orthonormality	1.431±0.840	0.160±0.066†	0.765±0.289†	0.733±0.261†
Staring affinity	1.903±1.180	0.090±0.040†	0.942±0.339†	0.594±0.205†
Distortion short (%)	13.41±3.62	5.53±1.04†	8.76±1.04†	5.01±1.13†
Distortion long (%)	4.44±0.83	3.17±0.68†	3.53±0.63†	1.83±0.62†
Regularity				
NDV (ppm)	7.9±29.2	7.9±29.3†	0.7±0.6	0.6±0.6†
NDV in bone (ppm)	6.5±4.6	0.3±1.2†	16.5±23.6‡	9.6±23.7
NDV in bone band (ppm)	1.9±1.7	0.7±0.9†	3.7±4.0‡	3.6±4.6‡
Preservation				
|ΔMTV| (%)	8.2±10.1	7.1±8.7	7.9±10.0	8.0±8.9
|ΔTLG| (%)	10.3±10.2	9.2±9.4	9.6±10.2	10.5±9.4